Pooling exact power certificates for Diophantine consistency

A fixed quartic in 225 unrestricted integer variables

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Abstract

We construct a fixed polynomial of total degree four in 225 unrestricted integer variables whose integer unsolvability is provably equivalent in ZFC to Con(ZFC), by composing the new arithmetic proof with the repository's identified C222 refutation calibration. The polynomial is the sum of squares of 162 literal quadratic residuals; its collected expansion has 7,867 nonzero monomials and coefficient height 128,446. The inspected parent has 245 variables, 181 residuals, 5,304 monomials, and the same degree and height. The principal new construction replaces all power congruences sharing one base by two quadratic equations with two auxiliary integers, independently of the number of requests. A sum of squared errors, a signed Pell congruence, and the already available square frame prove simultaneous exactness, including exclusion of negative exponents at nonunit bases. The modulus itself need not be a coordinate. Eleven requests at one base and three at another save twelve variables. Cofinal strict ranges and a smaller beta base save five more; exact changes of auxiliary coordinates save three. The delivered equation encodes a numerical proof calculus directly and uses neither K215 nor Jones's universal polynomial. The result improves this explicit direct quartic family. It is not a global minimality theorem or a new end-to-end proof-assistant certificate.

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1 The equation, theorem, and comparison

Let $\mathbf{x} = (x_0, \ldots, x_{224}) \in \mathbb{Z}^{225}$. The equation is

$$\boxed{P_{225}(\mathbf{x}) = \sum_{j=0}^{161} R_j(\mathbf{x})^2 = 0.} \tag{1}$$

Appendix A prints every residual R_j, with every coefficient and coordinate index. The same literal specification is `equation/equation.json`; the collected polynomial is `equation/expanded_polynomial.json`. All coefficients are fixed integers. No factorial, binomial operator, variable exponent, unspecified parameter, or black-box universal polynomial remains in (1).

Theorem 1.1. *For the literal polynomial in (1), the arguments below, composed with the retained arithmetic and C222/ZF/ZFC refutation calibrations, prove*

$$\mathrm{ZFC} \vdash \Big(\forall \mathbf{x} \in \mathbb{Z}^{225},\ P_{225}(\mathbf{x}) \neq 0\Big) \quad \longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

The polynomial has exact total degree four, 7,867 collected nonzero monomials, and maximum absolute coefficient 128,446.

The consistency assertion is syntactic consistency for the effective ZFC presentation fixed by the inherited calibration. The construction assumes neither consistency nor inconsistency. Its coefficients and verification artifacts are obtained by finite arithmetic independently of whether a refutation exists.

The comparison is pinned to ProveIt commit

`a0ef2df5bb1f73c35cd334bbc7f53553350ef06a`.

All nine incoming archives at that snapshot were downloaded and their relevant reports inspected. The immediate baseline is the *Joint Norm* archive, whose 245-variable direct polynomial supersedes the older 252-variable endpoint for this comparison [1, 2]. The main collection's top-level summary does not yet include that incoming improvement. It would therefore be misleading to compare only with its older 282-variable direct result.

Measure	Inspected parent	This equation	Change
Unrestricted integer variables	245	225	−20
Quadratic residuals	181	162	−19
Total degree	4	4	0
Collected monomials	5,304	7,867	+2,563
Coefficient height	128,446	128,446	0
Largest residual coefficient	256	256	0

The reduction in variables is approximately 8.16%. The larger monomial count is a real tradeoff: squaring a pooled residual creates more cross terms. The claim is fewer variables at the same degree and coefficient height, not improvement of every complexity measure.

What "simplest" can mean. The repository also contains equations with far fewer variables but substantially larger degree or large fixed coefficient encodings [3]. Sun's ten-integer undecidability theorem concerns a different degree/variable tradeoff [6]; the present construction does not improve that general variable bound. Our objective here is a direct quartic with moderate literal coefficients and a transparent arithmetic certificate. We do not know an absolute simplest equation, and no literature-wide priority or minimality claim is made.

2 The new idea: two equations for many exact powers

The inherited construction already classifies a finite family of signed Pell pairs at one large parameter. Its remaining power checks had one congruence quotient per request and one modulus per base. Those separate quotients can be removed in groups. The important feature is that the aggregation remains quadratic. The earlier batched-certificate proposal shares the Pell environment while keeping individual power congruences [4]; the present pooling removes those individual quotients.

2.1 Signed Pell notation and the classical congruence

For an integer A with $|A| > 1$, define the Pell polynomials by

$$\chi_A(0) = 1, \qquad \psi_A(0) = 0,$$
$$\begin{pmatrix} \chi_A(k+1) \\ \psi_A(k+1) \end{pmatrix} = \begin{pmatrix} A & A^2-1 \\ 1 & A \end{pmatrix} \begin{pmatrix} \chi_A(k) \\ \psi_A(k) \end{pmatrix}.$$

The matrix has determinant one, so the definition extends to every $k \in \mathbb{Z}$, with $\chi_A(-k) = \chi_A(k)$ and $\psi_A(-k) = -\psi_A(k)$. In particular,

$$\chi_A(k)^2 - (A^2-1)\psi_A(k)^2 = 1.$$

These functions occur in the proof only. Existing quadratic equations certify their coordinates in the delivered polynomial.

For an integer base n, put

$$T = 2An - n^2 - 1. \tag{2}$$

Then $\gcd(n, T) = 1$, since $T \equiv -1 \pmod{n}$. The standard power congruence is

$$\chi_A(k) + (n-A)\psi_A(k) \equiv n^k \pmod{T}, \qquad k \in \mathbb{Z}, \tag{3}$$

where a negative exponent on the right is interpreted in the group of units of $\mathbb{Z}/(T)$. To prove it, substitute $n - A$ for the formal square root of $A^2 - 1$: $(n-A)^2 \equiv A^2 - 1 \pmod{T}$, and the fundamental unit maps to $A+(n-A) = n$. Positive powers follow by multiplication and negative powers by invertibility. This proves the signed version directly. Classical Diophantine power and factorial representations are also formalized in existing libraries and work such as [5]; our new step is the following aggregate use of the congruence.

2.2 An exact quadratic identity

Suppose a group has one common base n and requests (k_i, m_i), $1 \le i \le r$. Write

$$X_i = \chi_A(k_i), \qquad Y_i = \psi_A(k_i), \qquad Z_i = AY_i.$$

The Z_i are existing coordinates, with quadratic definitions. For the proof abbreviate

$$D_i = m_i - X_i + Z_i, \qquad H = \sum_{i=1}^{r} Y_i(m_i - X_i), \qquad S = \sum_{i=1}^{r} (D_i^2 - Y_i^2).$$

Introduce only two integers K, L, and impose

$$K - 2\sum_{i=1}^{r} Y_i(m_i - X_i) - (2A - n)L = 0, \tag{4}$$

$$\sum_{i=1}^{r} \big((m_i - X_i + Z_i)^2 - Y_i^2\big) - nK + L = 0. \tag{5}$$

Both residuals have total degree at most two. Neither H, S, D_i, nor T is an additional coordinate. The number of auxiliary variables is independent of r.

Lemma 2.1 (Exact aggregate identity)**.** *Assume $Z_i = AY_i$. Equations (4)–(5) imply*

$$\sum_{i=1}^{r}\bigl(m_i - X_i + (A-n)Y_i\bigr)^2 = T\left(L - \sum_{i=1}^{r} Y_i^2\right). \tag{6}$$

Conversely, if each individual power error on the left is divisible by $T \neq 0$, integers K, L satisfying the two quadratic rows exist.

Proof. Substitution of $K = 2H + (2A-n)L$ into the second row gives $S - 2nH = TL$. On the other hand,

$$\begin{aligned}\sum_i (D_i - nY_i)^2 &= \sum_i D_i^2 - 2nH - (2An - n^2)\sum_i Y_i^2 \\ &= S - 2nH - T\sum_i Y_i^2.\end{aligned}$$

This is (6). Conversely set

$$L = \sum_i Y_i^2 + \frac{\sum_i (m_i - X_i + (A-n)Y_i)^2}{T}, \qquad K = 2H + (2A-n)L.$$

The assumed divisibilities make L integral, and the identity proves both rows. No sign condition on either new integer is needed. □

One cannot infer that each error vanishes merely from $T \mid \sum e_i^2$. The next theorem supplies the essential small-error argument. It applies to the small semantic errors obtained after Pell reduction, not to the very large raw Pell coordinates.

2.3 The existing square frame is sufficient

The inherited frame is

$$W = 2 + \sum_{a\in\mathcal{A}} a^2, \qquad |A| > W^W, \tag{7}$$

where every actual base, requested exponent, and proposed output is an atom in $\mathcal{A}$. Repeated atoms cause no problem. In our application $W \geq 258$. Thus the stronger bounds $n^2, k_i^2, m_i^2 < W$, not just $|n|, |k_i|, |m_i| < W$, are available.

Theorem 2.2 (Quadratic pooling of signed powers)**.** *Suppose $W \geq 258$, $1 \leq r \leq 15$, $n \neq 0$,*

$$|A| > W^W, \qquad n^2, k_i^2, m_i^2 < W,$$

and the pairs (X_i, Y_i) are the canonical signed Pell pairs at the indices k_i, with $Z_i = AY_i$. Then the two quadratic equations (4)–(5) force

$$\begin{cases} k_i \geq 0 \text{ and } m_i = n^{k_i}, & |n| \geq 2, \\ m_i = n^{k_i}, & n \in \{-1, 1\}. \end{cases}$$

For unit bases the integer power is defined for every signed index. Conversely, these exact power values admit K, L.

Proof. Put $b = \lfloor \sqrt{W-1} \rfloor$. Then $b \geq 16$, $|n|, |k_i|, |m_i| \leq b$, and $b + k_i \geq 0$. By (6) and (3), multiplication by n^{2b} gives

$$T \mid E, \qquad E := \sum_{i=1}^{r} (m_i n^b - n^{b+k_i})^2.$$

This is an ordinary nonnegative integer: the exponent shift has removed every inverse. As $b \geq 2$,

$$|m_i n^b - n^{b+k_i}| \leq b^{b+1} + b^{2b} \leq 2b^{2b}.$$

Consequently,

$$0 \leq E \leq 4rb^{4b} < W^{2b+1} \leq W^W. \tag{8}$$

Here $4r \leq 60 < W$, $b^2 < W$, and $2b + 1 \leq b^2 + 1 \leq W$. Also

$$|T| \geq 2|A||n| - n^2 - 1 \geq 2|A| - W > |A| > W^W.$$

Thus $0 \leq E < |T|$ and $T \mid E$, so $E = 0$. Every summand vanishes. If $k_i \geq 0$, cancel the nonzero integer n^b to obtain $m_i = n^{k_i}$. If $k_i = -j < 0$, cancellation gives $m_i n^j = 1$. This is impossible for $|n| \geq 2$; for $n = \pm 1$ it gives exactly the asserted unit power. Conversely the classical congruence makes every individual error divisible by T, and Lemma 2.1 constructs K, L. □

The number fifteen is a convenient bound for this particular application. The identical proof works for any number r of requests when $4r < W$ and the other frame hypotheses hold. The two auxiliary coordinates do not grow with the number of requests.

The use of the square bounds is decisive. A coarser estimate using only $|k_i| < W$ would suggest enlarging the Pell frame. That enlargement is unnecessary here. The final polynomial retains the original odd Pell-frame certificate unchanged. The finite checker reconstructs the frame as a sum of 36 affine squares after the removed request and authenticates every pooled base, exponent, and output against that list.

2.4 The two actual groups and the variable saving

After the strict-horizon change below, the direct source has seventeen power requests. Fourteen belong to the two groups in this table. The symbols are semantic values, before the final integral coordinate changes.

Base	Requested indices	Number
$q + 1$	$F^2 + t, a, b, c, d, S, S', u, v, C, h$	11
q	$t, y + 1, x$	3
$F^2 + 1$	t	1
B	x	1
BH	x	1

The nine independent CRT representatives are $a, b, c, d, S, S', u, v, C$. The derived argument h is a fixed constructor polynomial in the four schema representatives. The last three requests remain individual, which makes the bootstrap simple. Each pooled group removes its old modulus coordinate and all its old quotients, and introduces two integers. The savings are therefore

$$(11 + 1) - 2 = 10, \qquad (3 + 1) - 2 = 2.$$

The existing Pell pairs and their multiplication coordinates are retained. The theorem is not using unrecorded powers as primitives; the entire certificate consists of the retained quadratic rows and the four new quadratic rows.

3 Cofinal horizons remove a complete power request

The horizon is selected existentially from $t = 16 + z^2$. A finite refutation permits arbitrarily large choices of this form before the factorial and accumulator states are constructed. This freedom makes weak upper bounds unnecessarily expensive.

3.1 Strict local ranges

The parent uses binomial divisibilities with lower index $t + 1$ for four local schema parameters and one locally decoded derived-label sum. Use lower index t instead. They now enforce

$$0 \le a, b, c, d < t, \qquad 0 \le J(a, b, c, d) < t$$

on every decoded local tuple. All inherited soundness arguments require only the weaker bounds $\le t$, so the stronger conditions suffice. The one-sided prime-power binomial range lemma gives this conclusion in exactly the same manner as in the parent.

The notation must be kept straight. The local value $J(a, b, c, d)$ is bounded by t. The value $h = J(A_a, A_b, A_c, A_d)$ on the large global CRT representatives is generally much larger and is the semantic Pell anchor. No bound $h < t$ is imposed.

For completeness, take a finite proof tree, list all its parameter tuples, and add the fixed truth-padding tuple. Choose $t = 16 + z^2$ strictly larger than its number of expansion steps, every parameter, and every full derived-label sum on that finite list. These sums are independent of $F = t!$, so this choice is not circular.

At the parent's precision $2x$, a global representative is congruent to the intended local value $w < t$ modulo p^{2xm}. If the global value is smaller than t, the required binomial is zero. Otherwise its falling factorial has a factor congruent to zero modulo p^{2xm}; division by $t!$ loses at most $t - 1 < xm$ powers of p. Thus the required p^{xm} divisibility holds. Polynomial congruence proves the same statement for the derived-label value.

3.2 One denominator disappears

Put $M = q^x$. The parent normalizes digit remainders by ratios M/q^t, M/q^{t+1}, and M/q^{y+1}, with a fourth group already using M. After tightening the five ranges, every use of M/q^{t+1} can use M/q^t. Delete the output q^{t+1}, its power quotient, the private wire for its one-step Pell addition, and its normalization ratio. This saves four variables and three rows.

The removed output's square is also deleted from the square frame. Every retained base, exponent, and output remains bounded by an explicit square. The unchanged t pair is still needed by both q^t and $(F^2 + 1)^t$. The modified early bootstrap obtains $x \ge t$, rather than $x \ge t + 1$; Section 5 proves the adjusted estimates explicitly.

4 A smaller beta base

Let $V = (F^2 + 1)^t$. Replace the parent's $y = V(V - 1)$ by the affine identity

$$y = V - 1. \tag{9}$$

This is another use of the cofinal horizon, and eliminates one coordinate.

Starting with one pending formula, each backward expansion of a proof tree replaces one formula by at most two premises. If the finite tree requires L expansions, every pending multiset has at most $L + 1 \le t$ elements. Truth padding adds only a weight-one operation and does not enlarge the nontrivial

state. Every formula label j lies below t, and the inherited multiplicative weight is $1 + jF$. Since $F = t! \geq t$,

$$1 + jF \leq 1 + (t-1)F < F^2 + 1.$$

Every positive accumulator state is therefore strictly below $(F^2+1)^t = V$, and at most y. Also $F \mid V - 1 = y$. This is exactly the divisibility needed to make the beta moduli $1 + (i+1)y$, for $i = 0, \ldots, t$, pairwise coprime. Indeed a common prime divisor would divide an index difference, hence F, hence y, which is incompatible with dividing $1 + (i+1)y$.

After factorial recovery, $y > F + 2$ still holds by an enormous margin. Thus the endpoint and amplification arguments retain their hypotheses. Before that recovery, $V \geq 1$ gives $y \geq 0$, including the temporary branch $F = 0, V = 1, y = 0$, which is separately excluded below.

There is an offset in the centered parent coordinates: actual $y = x_2 - 1$, whereas $V = x_6$. The literal new row is consequently $x_2 - x_6 = 0$, not $x_2 - x_6 + 1 = 0$. Its exact elimination is checked in the supplied certificate. This detail matters for both endpoints.

5 Soundness in an acyclic order

We give the dependency order in some detail because signed witnesses and the deleted initial guards make premature positivity assumptions unsafe. The argument retains the parent's rounded factorial certificate and shared remainder norm; the changed pieces are stated explicitly.

5.1 What is available first

The retained signed Pell cap and pooled coordinate-orientation argument give all fourteen principal pairs at their requested signed indices, before interpreting any front power. They use the square frame, a separate squared semantic anchor, and bounded residues modulo $A - 1$. Their hypotheses are unchanged by pooling; the cap does not depend on individual power quotients. The retained proof texts are listed in Section 8.

Among the three individual power requests, the request at $F^2 + 1$ first gives $V = (F^2+1)^t$, since its index $t \geq 16$ is already known. The shared norm has the semantic form

$$M^2 = 1 + 4\left(e^2 + \sum_{j=1}^{11} \rho_j^2\right) + \mathcal{G}(\alpha, \beta, \gamma), \qquad \mathcal{G}(u, v, w) = u^2 + v^2 + w(w+1), \tag{10}$$

and the ceiling rows are

$$Bd = M, \qquad FM - dV - e = 0. \tag{11}$$

Here $\mathcal{G} \geq 0$ for all integer inputs. Thus $M, B, d \neq 0$, and $e = d(FB - V)$ gives

$$|FB - V| < |B|/2. \tag{12}$$

Write $b_0 = |B|$, $f = |F|$. It follows that $1 \leq V < (f + 1/2)b_0$. If $f = 0$, then $b_0 > 2$. If $f \geq 1$ and $b_0 \leq f$, then $(f^2+1)^t \geq (f^2+1)^2 > f^2 + f/2$, a contradiction. Hence

$$b_0 \geq 2, \qquad |F| < b_0, \qquad V < b_0^2. \tag{13}$$

5.2 The unit-base case licenses the first pooled group

Set $H = 1 + 2\sum a_i$, where the sum runs over the nine actual CRT representatives. It is odd, hence nonzero, without any sign premise. The individual requests at B and BH force

$$x \geq 0, \qquad N = B^x, \qquad q = (BH)^x.$$

In particular $q \neq 0$, so Theorem 2.2 applies to the three requests at base q, even if temporarily $q = 1$. If $x = 0$, then $q = 1$, and the pooled request for $M = q^x$ gives $M = 1$. This contradicts $Bd = M$ and $|B| \geq 2$. Therefore $x \geq 1$ and $|q| \geq 2$.

The retained denominator identity $q^t r_t = q^x$ now gives $x \geq t$: a negative power of an integer of absolute value at least two cannot be an integer ratio. Consequently

$$|q| \geq b_0^x \geq 2^{16}, \qquad |q+1| \geq 2^{16} - 1 \geq 2.$$

The eleven-request group at base $q + 1$ now forces every actual representative to be nonnegative and all its powers to be exact. It also forces the derived requested index to be nonnegative; the independent anchor has already excluded zero. Thus $H \geq 1$.

This order explains why the theorem's unit-base clause matters. Demanding $|q| \geq 2$ as a premise before proving $x \neq 0$ would leave a gap.

5.3 Factorial digit recovery, including the zero candidate

Let $Q = |q|$. The binomial numerator is now the exact signed-radix power $(q+1)^{F^2+t}$. If $F \neq 0$, each coefficient through degree t is at most $V = (F^2+1)^t$. The signed low tail r^0 at position t therefore satisfies

$$\frac{|r^0|}{Q^t} < \frac{V}{Q-1} < \frac{2}{b_0^{x-2}} \leq \frac{2}{2^{14}} < \frac{1}{32}.$$

The common norm bounds the absolute value of every chosen remainder by half its absolute modulus. Two congruent remainders, one of absolute value less than half that modulus and the canonical one of absolute value less than one thirty-second of it, cannot differ by a nonzero multiple of the modulus. They are equal. The next digit gives

$$B \equiv \binom{F^2+t}{t} \pmod{q}.$$

Both values have difference less than $b_0 + b_0^2 < Q$, so $B = \binom{F^2+t}{t} > 0$.

When $F = 0$, it would be wrong to bound all $\binom{t}{j}$ by one. Instead $V = 1$, (12) gives $b_0 \geq 3$, and

$$\frac{|r^0|}{Q^t} < \frac{2^t}{Q-1} < 2(2/3)^t \leq 2(2/3)^{16} < \frac{1}{32}.$$

The same recovery forces $B = \binom{t}{t} = 1$, contradicting $b_0 \geq 3$. Thus this branch has no zero.

5.4 The rounded factorial lemma

For completeness we recall the inherited elementary fixed-point argument on the range needed here. If $t \geq 16$, $F \in \mathbb{Z}$,

$$B = \binom{F^2+t}{t}, \qquad V = (F^2+1)^t,$$

then $|F - V/B| < 1/2$ holds exactly when $F = t!$. Put $L_0 = t!$, $a = t(t-1)/2$, and

$$f_t(u) = L_0 \prod_{j=1}^{t} \frac{u^2+1}{u^2+j}.$$

This is nondecreasing for $u \geq 0$, lies in $(0, L_0)$, and

$$f_t(u) \geq L_0 \left(1 - \frac{a}{u^2+1}\right).$$

The half-width condition excludes $F \le 0$ and $F \ge L_0{+}1$. For $1 \le F \le t{-}1$, $f_t(F) \ge 2^t/(t{+}1) > t{-}1/2 \ge F + 1/2$. For $t \le F \le L_0/2$, the displayed lower bound exceeds $L_0/2 + 1/2$. For $L_0/2 \le F \le L_0 - 1$, it gives $L_0 - f_t(F) < 4a/L_0 = 2/(t-2)! < 1/2$, again excluding F. At $F = L_0$,

$$0 < L_0 - f_t(L_0) < \frac{a}{L_0} = \frac{1}{2(t-2)!} < \frac{1}{32},$$

which proves both the characterization and the completeness margin.

We have now recovered $F = t!$, $B > 0$, $q > 0$, and $y = V - 1 > F + 2$. The endpoint identities are

$$x = F + 1 + (y+1)u_0, \qquad x = 1 + \big(1 + (t+1)y\big)u_t.$$

Since $x \ge t > 1$, the second gives $u_t \ge 1$, hence

$$x \ge 2 + (t+1)y. \tag{14}$$

The inherited coefficient estimates recover all remaining digit ranges. The lowered positions in Section 3 only strengthen them. The strict amplification proof, using $R = 2 + t + x + y + F < 2x$, the bound $x > 2^{32}$, and the retained majorant exponent 2^{23}, again gives $R^{2^{23}} < 2^x$. The finite CRT and Horn-flow decoding therefore produce a C222 refutation.

6 Completeness from a finite refutation

Start with a finite C222 refutation. Select the strict cofinal horizon as in Section 3; then define $F = t!$, $V = (F^2+1)^t$, and $y = V - 1$. Section 4 supplies the pending-state bound and beta CRT hypotheses. The parent's construction gives the beta code x, the exact factorial-binomial value B, and the nine CRT representatives at precision N^2, where $N = B^x$.

Choose representatives cofinally in their residue classes, as in the retained semantic-anchor construction: the dominant schema representative can be between $2N^2$ and $4N^2$, with prescribed parity, while the others are at most $2N^2$. This makes the large global derived index h even and strictly larger than every other principal index. This choice does not alter the small local values or their strict bounds. Set $H = 1 + 2\sum a_i$, $q = (BH)^x$, and fill all seventeen genuine power values.

Only now form W from the semantic atoms. The ordinary parent frame chooses a sufficiently large canonical Pell abscissa at center $W + 1$, with ordinate an odd multiple of W. It gives $|A| > W^W$. The canonical principal pairs, squared semantic anchor, common auxiliary pair, and comparison pair exist by the retained constructive Pell and finite-period arguments. No new condition on the frame is imposed.

Every exact power satisfies its original power congruence, so Lemma 2.1 supplies the two aggregate integers for each group. Thus completeness of the new pooled rows follows from simultaneous completeness of the old congruences without requiring that their individual quotient coordinates remain in the equation.

Every true digit remainder, normalized to $M = q^x$, is smaller than $M/32$; the rounded factorial error has the same bound. The twelve errors therefore satisfy

$$4\left(e^2 + \sum_{j=1}^{11} \rho_j^2\right) < \frac{48}{1024} M^2 < M^2.$$

The remaining nonnegative integer slack is represented by $\mathcal{G}(u,v,w) = u^2 + v^2 + w(w+1)$. One quick proof of its universality is to apply the three-square theorem to $4n+1$: exactly one of the three square roots is odd, giving the claimed form after division by four. The other retained guard is filled in the same way.

Finally apply the exact coordinate maps of the next section. Every finite refutation thus yields an integer zero of the delivered polynomial. Together with soundness this proves the required equivalence of existential statements. The strict-horizon and beta changes go through freshly constructed refutation witnesses; they are not claimed to be bijections between arbitrary old and new assignments at a fixed horizon.

7 Exact reductions of auxiliary coordinates

Three further variables disappear by identities over $\mathbb{Z}$. These steps have explicit maps in both directions and do not need semantic bounds, positivity, or divisions.

7.1 Store a sum when only the sum is used

Suppose $u = f(\mathbf{z})$, $v = g(\mathbf{z})$ are private definitions, and every other residual uses these variables only through $u + v$. Replace them by $w = f(\mathbf{z}) + g(\mathbf{z})$. Projection gives $w = u + v$, and restoration gives $u = f$, $v = g$. If f, g are quadratic, their sum is still quadratic even when substituting either separately into a product would have raised degree.

In the centered parent, the two values are

$$x_{81} = x_{80}^2, \qquad x_{83} = (x_{14} - x_{15} - x_{82})^2.$$

Their only external use is $-x_{101}(x_{81} + x_{83})$. One sum coordinate therefore replaces the two and removes one variable and one row.

7.2 Compute the products that the remaining rows use

Let $F = x_3$, $S = x_{14}$, $S' = x_{15}$. The product $FS' = x_{70}$ already exists. Three auxiliary definitions in the centered parent are

$$x_{84} = 1 + Fx_{11}, \quad x_{85} = 1 + F(x_{10} - 1), \quad x_{89} = 1 + F(6x_{10} + 8x_{11} + 2x_{21}).$$

They are used only after multiplication by S or S'. Introduce one coordinate $J = FS$, delete these three definitions, and use the distributive identities

$$\begin{aligned} S'(1 + Fx_{11}) &= S' + (FS')x_{11}, \\ S(1 + F(x_{10} - 1)) &= S + J(x_{10} - 1), \\ S(1 + Fx_{11}) &= S + Jx_{11}, \\ S'(1 + F\ell) &= S' + (FS')\ell, \qquad \ell = 6x_{10} + 8x_{11} + 2x_{21}. \end{aligned}$$

All replacement residuals remain quadratic. The old values restore by their three definitions, so this saves two variables and two rows with an exact integral inverse on zero sets.

7.3 Pell difference coordinates reduce density

To control the density caused by pooling, replace each principal abscissa by the difference $D_i = X_i - Z_i$. This is a triangular integral change of coordinates, with inverse $X_i = D_i + Z_i$. The norm becomes

$$D_i^2 + 2D_iZ_i + Y_i^2 - 1 = 0,$$

and the long pooled square uses $(m_i - D_i)^2$. Before that substitution, reduce the orientation row by the retained norms. Modulo those norms its summand satisfies

$$(X_i - 1)^2 + (Y_i - k_i)^2 \equiv Z_i^2 + k_i^2 - 2X_i - 2k_iY_i + 2.$$

After the difference substitution, D_i occurs only linearly here. Adding or subtracting a retained residual from another is reversible and preserves the common zero set. The independent checker authenticates the exact row combinations and every offset in the centered coordinates.

These transformations yield the following total saving:

$$\underbrace{4}_{\text{strict ranges}} + \underbrace{1}_{\text{beta base}} + \underbrace{3}_{\text{auxiliary coordinates}} + \underbrace{12}_{\text{power groups}} = 20.$$

The last density change does not alter the number of variables.

8 Calibration and the formalization boundary

The underlying source is the numerical proof calculus C222, under the `xx0` square-shell recoding used by the direct arithmetic construction. The constructors, the eight ordinary axiom families, the guarded bundle, modus ponens, and generalization are retained exactly from the immediate parent. Falsity has code 1 and truth code 0. The new equation changes how finite refutation certificates are represented arithmetically, not which calculus is calibrated.

The semantic chain is

$$\begin{aligned} \exists \mathbf{x}\ P_{225}(\mathbf{x}) = 0 \quad &\longleftrightarrow \quad \text{a finite square-shell C222 derivation of 1} \\ &\longleftrightarrow \quad \text{a finite original C222 derivation of 1} \\ &\longleftrightarrow \quad \text{a finite ZF refutation} \\ &\longleftrightarrow \quad \text{a finite ZFC refutation.} \end{aligned}$$

For clarity, the recoding is an ordinary theorem rather than a claim that the two sets of constructor polynomials are identical. Put

$$Q_0(a,b) = (a+b)^2 + a, \qquad Q_1(a,b) = (a+b)^2 + 3a + 4b + 3.$$

The shell constructors are $I = 2Q_0$, $M = 2Q_0 + 1$, $\mathsf{A} = 2Q_1$, and $\mathsf{E} = 2Q_1 + 1$. On the shell $a + b = s$, Q_0 covers $[s^2, s^2 + s]$ once; on the preceding shell, Q_1 covers $[s^2 + s + 1, (s+1)^2 - 1]$ once. These intervals partition $\mathbb{N}$. Every positive constructor value exceeds both arguments. Decoding and recursing therefore gives a total constructor-preserving bijection f from the original Cantor coding to this shell coding, with the shared case $I(0,0) = 0$ treated as a base case. Its inverse is obtained by the opposite decoding; induction proves both inverse identities. In particular $f(0) = 0$, $f(1) = 1$, and equality and disequality are preserved.

All constructor-term axioms and the two inference rules transport by f. In the guarded branch, the arithmetic witness must be rebuilt: from the two transported freshness disequalities choose

$$d_{\text{new}} = |(f(c) - f(a))(f(c) - f(b))| - 1 \geq 0.$$

This gives the required squared guard. It does not assume that f preserves addition. A separate branch in which d is a code parameter transports d normally; the branches are alternatives, so these assignments do not conflict. The inverse bijection proves the converse transport. The full inherited argument is supplied unmodified in `inherited/shell_recoding_supplement/SHELL_ENCODING_PROOF.md`, with the literal constructor sources and their pinned provenance.

The named logical interfaces in the retained source are `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. The latter uses the constructible-universe equiconsistency argument. The relevant interface files, source pins, ordinary arithmetic proofs, and original notices are included under `inherited/joint_norm245/`; the separately identified syntax

supplement closes the recoding step. These retained sources are not a complete recompiled proof-assistant dependency closure.

For auditing the new arithmetic, the most relevant retained proof files are the Joint Norm `proofs/BOOTSTRAP_PROOF.md` and, within its normalized-parent closure, `PROOF_315.md`, `ORIENTATION_FIRST_PROOF.md`, `STRICT_BUDGET_PROOF.md`, `INHERITED_DERIVED_LABEL_RANGE_PROOF.md`, and the direct calculus and CRT exposition. The provenance manifest gives exact file paths and hashes, rather than relying on these abbreviated names to identify a dependency.

All new arguments use integer polynomial identities, induction on finite recurrences and proofs, elementary bounds, finite CRT, and finite-group Pell periodicity. They are ordinary mathematics formalizable in ZFC. Together with the identified inherited calibration they prove Theorem 1.1. The supplied Python checks certify finite objects and exact transformations. They do not certify the absence of zeros, establish consistency, or replace the unbounded mathematical proofs. The full imported Lean dependency closure was not recompiled in this work, and no new end-to-end Lean or Rocq theorem is claimed.

9 Reproduction and finite evidence

The package contains the literal residual system, its full sparse expansion, the construction stages, transformation certificates, the unmodified immediate parent and retained proof closure, the report source, and independent verification scripts. Python 3 with its standard library is sufficient for the new finite checks.

From the package root, run

```
python3 -B verify.py
python3 -B -O verify.py
```

The build command and the PDF rebuild command are in `README.md`. The new verifier checks source identities, the semantic edit boundaries, the auxiliary-coordinate maps, the two power groups and their exact affine ports, the unchanged square frame and atom coverage, the Pell coordinate changes, and every coefficient of the collected polynomial. It computes the reported dimensions, degree, support, coefficient height, and coordinate liveness from the delivered residuals.

Separate exact checks exercise the aggregate polynomial identity, signed Pell congruences, unit and negative bases, negative indices, the square-frame error estimate, and the corrected zero-factorial bound. These are useful falsification and regression checks; the preceding theorems establish the unbounded conclusions. The reviews were independently assigned to other AI research agents, not represented as external human peer review.

10 Further research questions

1. **Can distinct bases share one aggregate?** The present proof groups by an identical base and uses its single modulus $2An - n^2 - 1$. Pooling different bases without new product or CRT coordinates would require a different quadratic identity. A naive sum of their errors has no common modulus and is not justified by this proof.

2. **Can an entire Pell pair disappear?** We removed quotient coordinates while keeping canonical pairs. A useful next target is a semantic exponent identity that reconstructs a pair already needed elsewhere, with no new multiplication coordinates.

3. **How far can auxiliary-coordinate compression go?** The sum replacement and product redistribution exploit how values are used by the remaining equations. A systematic search

over integral changes of coordinates and quadratic row combinations may find more reductions than single-row monic elimination. Coefficient height and support must be measured alongside dimension.

4. **Can the new module be formally verified independently?** The aggregate identity, signed modular power lemma, and square-frame error bound form a small coherent target for Lean or Rocq. A theorem with an explicit list of affine ports would allow subsequent compilations to reuse one checked module.

5. **What is the best calculus for the arithmetic cost?** Optimizing the number of schema parameters can help, but a parameter hidden in a marker premise may reappear as an additional encoded column in the global proof-flow construction. A local reduction must be carried through the complete arithmetic compiler before claiming a smaller consistency equation. The full guarded-bundle freshness range must also be retained unless a new logical calibration is proved.

6. **Can density be reduced without giving back the variables?** The principal equation has more collected monomials than its parent. Alternative quadratic residual bases and integral Pell coordinates are natural targets for a future support/height optimization at dimension 225.

Attribution

The C222 calibration, finite flow/CRT encoding, rounded factorial, signed Pell cap, cofinal semantic anchor, and common digit norm are inherited and identified in the source ledger. The new contributions are the two-equation pooling of exact powers at a common base with no modulus coordinate, its signed square-frame proof and integration into the bootstrap, the strict-horizon and linear beta refinements, the exact auxiliary-coordinate reductions, and the literal combined endpoint. No claim is made that elementary polynomial identities themselves are new. Novelty is stated relative to the inspected constructions; a global priority claim would require a broader literature review.

References

[1] V. Reshetnikov, *ProveIt*, snapshot `a0ef2df5bb1f73c35cd334bbc7f53553350ef06a`, 25 October 2026. https://github.com/VladimirReshetnikov/ProveIt/commit/a0ef2df5bb1f73c35cd334bbc7f53553350ef06a.

[2] *A single norm for factorial and digit certificates*, incoming package `ConZFC_Joint_Norm_2026-10-25.zip`, in the preceding repository snapshot. The unmodified direct component is included here. https://github.com/VladimirReshetnikov/ProveIt/blob/a0ef2df5bb1f73c35cd334bbc7f53553350ef06a/docs/incoming/ConZFC_Joint_Norm_2026-10-25.zip.

[3] V. Reshetnikov, *Con(ZFC) in Diophantine form*, collection README and placement ledger, same snapshot. https://github.com/VladimirReshetnikov/ProveIt/tree/a0ef2df5bb1f73c35cd334bbc7f53553350ef06a/Computability/DiophantineConsistency/con-zfc-diophantine-forms.

[4] *Shared arithmetic certificates for Diophantine consistency equations*, collection package 96, especially `report/batched_powers.tex`. This earlier batching shares the Pell environment and retains separate local power congruences.

[5] K. Pąk and C. Kaliszyk, *Formalizing a Diophantine Representation of the Set of Prime Numbers*, ITP 2022, arXiv:2204.12311. https://arxiv.org/abs/2204.12311.

[6] Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594v2, 2026. https://arxiv.org/abs/2609.01594v2.

A Literal residual specification

The following 162 residuals, in precisely this order, define (1). All x_i range independently over $\mathbb{Z}$. A line break within a residual does not start a new equation. This appendix is generated directly from `equation/equation.json` and checked against that file. The variable names in the machine-readable list are provenance labels; only the displayed integer coefficients and indices define the polynomial. Every index from 0 through 224 occurs.

$$R_0 = -x_2^2 + x_3$$

$$R_1 = x_0 - x_0x_5 - x_5 + x_8$$

$$R_2 = -1 + x_1 - x_2 - x_5x_6$$

$$R_3 = -1 + x_1 - x_7x_8$$

$$R_4 = 1 + x_3 - x_4x_{19} + x_{17}$$

$$R_5 = -1 + 2x_9 - x_9^2 - 2x_9x_{10} + 2x_{10} - x_{10}^2 + x_{20}$$

$$\begin{aligned} R_6 = {} & -5 - 30x_9 - 36x_9^2 - 96x_9x_{10} - 24x_9x_{20} - 40x_{10} - 64x_{10}^2 - 32x_{10}x_{20} - 10x_{20} \\ & - 4x_{20}^2 + x_{21} \end{aligned}$$

$$R_7 = -2 + 2x_9 - x_9^2 - 4x_9x_{21} + 4x_{21} - 4x_{21}^2 + x_{22}$$

$$\begin{aligned} R_8 = {} & 1 + 6x_9 - 36x_9^2 - 192x_9x_{21} - 24x_9x_{22} + 16x_{21} - 256x_{21}^2 - 64x_{21}x_{22} + 2x_{22} \\ & - 4x_{22}^2 + x_{23} \end{aligned}$$

$$R_9 = -2 - x_{10}^2 - 2x_{10}x_{11} - x_{11}^2 + x_{24}$$

$$\begin{aligned} R_{10} = {} & -9 + 7x_9 - x_9^2 - 4x_9x_{10} - 4x_9x_{24} + 12x_{10} - 4x_{10}^2 - 8x_{10}x_{24} + 12x_{24} - 4x_{24}^2 \\ & + x_{25} \end{aligned}$$

$$R_{11} = -1 + 2x_9 - x_9^2 - 2x_9x_{11} + 2x_{11} - x_{11}^2 + x_{26}$$

$$\begin{aligned} R_{12} = {} & -6x_9 - 144x_9^2 - 192x_9x_{10} - 192x_9x_{11} - 48x_9x_{20} - 48x_9x_{26} - 8x_{10} - 64x_{10}^2 \\ & - 128x_{10}x_{11} - 32x_{10}x_{20} - 32x_{10}x_{26} - 64x_{11}^2 - 32x_{11}x_{20} - 32x_{11}x_{26} - 2x_{20} \\ & - 4x_{20}^2 - 8x_{20}x_{26} - 4x_{26}^2 + x_{27} \end{aligned}$$

$$R_{13} = -4x_{25}^2 - 8x_{25}x_{27} - 4x_{27}^2 + x_{28}$$

$$\begin{aligned} R_{14} = {} & -6 + 16x_{11} - 16x_{11}^2 - 16x_{11}x_{24} - 16x_{11}x_{26} + 8x_{24} - 4x_{24}^2 - 8x_{24}x_{26} + 8x_{26} \\ & - 4x_{26}^2 + x_{29} \end{aligned}$$

$$R_{15} = 1 + 6x_9 - 36x_9^2 - 96x_9x_{10} - 48x_9x_{11} - 24x_9x_{20} - 48x_9x_{26} - 24x_9x_{29} + 8x_{10} - 64x_{10}^2 - 64x_{10}x_{11} - 32x_{10}x_{20} - 64x_{10}x_{26} - 32x_{10}x_{29} + 8x_{11} - 16x_{11}^2 - 16x_{11}x_{20} - 32x_{11}x_{26} - 16x_{11}x_{29} + 2x_{20} - 4x_{20}^2 - 16x_{20}x_{26} - 8x_{20}x_{29} + 8x_{26} - 16x_{26}^2 - 16x_{26}x_{29} + 4x_{29} - 4x_{29}^2 + x_{30}$$

$$R_{16} = -10 - 5x_9 - x_9^2 - 12x_9x_{10} - 16x_9x_{11} - 4x_9x_{24} - 36x_{10} - 36x_{10}^2 - 96x_{10}x_{11} - 24x_{10}x_{24} - 48x_{11} - 64x_{11}^2 - 32x_{11}x_{24} - 12x_{24} - 4x_{24}^2 + x_{31}$$

$$R_{17} = -2 - 24x_9 - 36x_9^2 - 12x_9x_{10} - 96x_9x_{11} - 24x_9x_{26} - 3x_{10} - x_{10}^2 - 16x_{10}x_{11} - 4x_{10}x_{26} - 32x_{11} - 64x_{11}^2 - 32x_{11}x_{26} - 8x_{26} - 4x_{26}^2 + x_{32}$$

$$R_{18} = 2 - 4x_{31}^2 - 8x_{31}x_{32} - 4x_{32}^2 + x_{33}$$

$$R_{19} = -12 + 16x_9 - 4x_9^2 - 8x_9x_{11} - 8x_9x_{20} - 8x_9x_{24} + 16x_{11} - 4x_{11}^2 - 8x_{11}x_{20} - 8x_{11}x_{24} + 16x_{20} - 4x_{20}^2 - 8x_{20}x_{24} + 16x_{24} - 4x_{24}^2 + x_{34}$$

$$R_{20} = -3 - 24x_9 - 64x_9^2 - 160x_9x_{11} - 64x_9x_{24} - 32x_9x_{26} - 32x_9x_{34} - 26x_{11} - 100x_{11}^2 - 80x_{11}x_{24} - 40x_{11}x_{26} - 40x_{11}x_{34} - 8x_{24} - 16x_{24}^2 - 16x_{24}x_{26} - 16x_{24}x_{34} - 6x_{26} - 4x_{26}^2 - 8x_{26}x_{34} - 4x_{34} - 4x_{34}^2 + x_{35}$$

$$R_{21} = -1 - 18x_9 - 36x_9^2 - 96x_9x_{10} - 24x_9x_{20} - 24x_{10} - 64x_{10}^2 - 32x_{10}x_{20} - 6x_{20} - 4x_{20}^2 + x_{36}$$

$$R_{22} = -2 + 2x_9 - x_9^2 - 4x_9x_{36} + 4x_{36} - 4x_{36}^2 + x_{37}$$

$$R_{23} = 1 + 6x_9 - 36x_9^2 - 192x_9x_{36} - 24x_9x_{37} + 16x_{36} - 256x_{36}^2 - 64x_{36}x_{37} + 2x_{37} - 4x_{37}^2 + x_{38}$$

$$R_{24} = -x_{10}^2 - 4x_{10}x_{38} - 4x_{38}^2 + x_{39}$$

$$R_{25} = -6 - 30x_9 - 64x_9^2 - 128x_9x_{10} - 64x_9x_{20} - 32x_{10} - 64x_{10}^2 - 64x_{10}x_{20} - 14x_{20} - 16x_{20}^2 + x_{40}$$

$$R_{26} = -4 + 8x_{10} - 64x_{10}^2 - 96x_{10}x_{11} - 32x_{10}x_{24} - 32x_{10}x_{40} + 6x_{11} - 36x_{11}^2 - 24x_{11}x_{24} - 24x_{11}x_{40} + 2x_{24} - 4x_{24}^2 - 8x_{24}x_{40} + 4x_{40} - 4x_{40}^2 + x_{41}$$

$$R_{27} = 8x_9 - 64x_9^2 - 96x_9x_{11} - 32x_9x_{26} - 32x_9x_{41} + 6x_{11} - 36x_{11}^2 - 24x_{11}x_{26} - 24x_{11}x_{41} + 2x_{26} - 4x_{26}^2 - 8x_{26}x_{41} + 4x_{41} - 4x_{41}^2 + x_{42}$$

$$R_{28} = -x_{11}^2 - 2x_{11}x_{12} - x_{12}^2 + x_{43}$$

$$R_{29} = 2x_9 - x_9^2 - 4x_9x_{11} - 4x_9x_{43} + 4x_{11} - 4x_{11}^2 - 8x_{11}x_{43} + 4x_{43} - 4x_{43}^2 + x_{44}$$

$$R_{30} = -35 + 24x_9 - 4x_9^2 - 8x_9x_{10} - 8x_9x_{24} - 8x_9x_{26} + 24x_{10} - 4x_{10}^2 - 8x_{10}x_{24} - 8x_{10}x_{26} + 24x_{24} - 4x_{24}^2 - 8x_{24}x_{26} + 24x_{26} - 4x_{26}^2 + x_{45}$$

$$R_{31} = -16x_9^2 - 32x_9x_{11} - 32x_9x_{26} - 32x_9x_{43} - 16x_9x_{44} - 16x_9x_{45} - 4x_{11} - 16x_{11}^2 - 32x_{11}x_{26} - 32x_{11}x_{43} - 16x_{11}x_{44} - 16x_{11}x_{45} - 16x_{26}^2 - 32x_{26}x_{43} - 16x_{26}x_{44} - 16x_{26}x_{45} - 4x_{43} - 16x_{43}^2 - 16x_{43}x_{44} - 16x_{43}x_{45} - 2x_{44} - 4x_{44}^2 - 8x_{44}x_{45} - 4x_{45}^2 + x_{46}$$

$$R_{32} = x_{11} - 25x_{11}^2 - 40x_{11}x_{26} - 20x_{11}x_{29} - 16x_{26}^2 - 16x_{26}x_{29} - 4x_{29}^2 + x_{47}$$

$$R_{33} = -1 + 8x_{10} - 4x_{10}^2 - 8x_{10}x_{24} + 8x_{24} - 4x_{24}^2 + x_{48}$$

$$R_{34} = -1 - 16x_{10}^2 - 8x_{10}x_{11} - 32x_{10}x_{24} - 16x_{10}x_{48} - x_{11}^2 - 8x_{11}x_{24} - 4x_{11}x_{48} - 16x_{24}^2 - 16x_{24}x_{48} - 4x_{48}^2 + x_{49}$$

$$R_{35} = 1 + x_{10} - x_{10}^2 - 4x_{10}x_{11} - 4x_{10}x_{49} - 4x_{11}^2 - 8x_{11}x_{49} - 4x_{49}^2 + x_{50}$$

$$R_{36} = -1 - 4x_{50} - 4x_{50}^2 + x_{51}$$

$$R_{37} = -1 - x_{11}^2 - 8x_{11}x_{50} - 4x_{11}x_{51} - 16x_{50}^2 - 16x_{50}x_{51} - 4x_{51}^2 + x_{52}$$

$$R_{38} = 2 + x_{10} - x_{10}^2 - 4x_{10}x_{11} - 4x_{10}x_{52} - 4x_{11}^2 - 8x_{11}x_{52} - 4x_{52}^2 + x_{53}$$

$$R_{39} = -4x_{47}^2 - 8x_{47}x_{53} - 4x_{53}^2 + x_{54}$$

$$R_{40} = -9 + 7x_9 - x_9^2 - 8x_9x_{47} - 4x_9x_{54} + 24x_{47} - 16x_{47}^2 - 16x_{47}x_{54} + 12x_{54} - 4x_{54}^2 + x_{55}$$

$$R_{41} = -1 - 4x_{55} - 4x_{55}^2 + x_{56}$$

$$R_{42} = -x_{11}^2 - 8x_{11}x_{55} - 4x_{11}x_{56} - 16x_{55}^2 - 16x_{55}x_{56} - 4x_{56}^2 + x_{57}$$

$$R_{43} = -x_9 - x_9^2 - 4x_9x_{11} - 4x_9x_{57} - 4x_{11} - 4x_{11}^2 - 8x_{11}x_{57} - 4x_{57} - 4x_{57}^2 + x_{58}$$

$$R_{44} = -10 - 12x_{58} - 4x_{58}^2 + x_{59}$$

$$R_{45} = 1 + 4x_{10} - 4x_{10}^2 - 8x_{10}x_{24} - 16x_{10}x_{58} - 8x_{10}x_{59} + 4x_{24} - 4x_{24}^2 - 16x_{24}x_{58} - 8x_{24}x_{59} + 8x_{58} - 16x_{58}^2 - 16x_{58}x_{59} + 4x_{59} - 4x_{59}^2 + x_{60}$$

$$R_{46} = x_{10} - 25x_{10}^2 - 40x_{10}x_{24} - 20x_{10}x_{60} - 16x_{24}^2 - 16x_{24}x_{60} - 4x_{60}^2 + x_{61}$$

$$R_{47} = -10 - 12x_{53} - 4x_{53}^2 + x_{62}$$

$$R_{48} = 1 - 16x_{53}^2 - 16x_{53}x_{61} - 16x_{53}x_{62} - 4x_{61}^2 - 8x_{61}x_{62} - 4x_{62}^2 + x_{63}$$

$$R_{49} = x_{11} - x_{11}^2 - 8x_{11}x_{61} - 4x_{11}x_{63} - 16x_{61}^2 - 16x_{61}x_{63} - 4x_{63}^2 + x_{64}$$

$$R_{50} = -4x_9 - 4x_9^2 - 4x_9x_{11} - 8x_9x_{20} - x_{11} - x_{11}^2 - 4x_{11}x_{20} - 4x_{20} - 4x_{20}^2 + x_{65}$$

$$R_{51} = 1 + 4x_9 - 4x_9^2 - 8x_9x_{20} - 8x_9x_{65} + 4x_{20} - 4x_{20}^2 - 8x_{20}x_{65} + 4x_{65} - 4x_{65}^2 + x_{66}$$

$$R_{52} = -2 - 4x_9 - 16x_9^2 - 32x_9x_{20} - 16x_9x_{64} - 16x_9x_{66} - 4x_{20} - 16x_{20}^2 - 16x_{20}x_{64} - 16x_{20}x_{66} - 4x_{64}^2 - 8x_{64}x_{66} - 2x_{66} - 4x_{66}^2 + x_{67}$$

$$R_{53} = -2 - 4x_{67} - 4x_{67}^2 + x_{68}$$

$$R_{54} = -x_2x_{14} + x_{69}$$

$$R_{55} = x_{13} - x_{14} - 2x_{23}x_{69} - x_{70}$$

$$R_{56} = x_{13} - x_{14} - 4x_{25}x_{69} - 2x_{28}x_{69} - x_{71}$$

$$R_{57} = x_{13} - x_{14} - 2x_{30}x_{69} - 6x_{69} - x_{72}$$

$$R_{58} = x_{13} - x_{14} - 4x_{31}x_{69} - 2x_{33}x_{69} - x_{73}$$

$$R_{59} = x_{13} - x_{14} - 2x_{35}x_{69} + 6x_{69} - x_{74}$$

$$R_{60} = x_{13} - x_{14} - 4x_{38}x_{69} - 2x_{39}x_{69} - x_{75}$$

$$R_{61} = x_{13} - x_{14} - 2x_{42}x_{69} - x_{76}$$

$$R_{62} = x_{13} - x_{14} - 2x_{46}x_{69} - 4x_{69} - x_{77}$$

$$R_{63} = -x_9x_{10} + x_9x_{11} + x_{10} + x_{10}x_{11} - x_{11} - x_{11}^2 + x_{78}$$

$$R_{64} = 1 + 2x_{12} + x_{12}^2 - x_{78}^2 + x_{79}$$

$$R_{65} = -x_{13}^2 + 2x_{13}x_{14} + 2x_{13}x_{81} - x_{14}^2 - 2x_{14}x_{81} - x_{79}^2 + x_{80} - x_{81}^2$$

$$R_{66} = -4x_{67}x_{69} - 2x_{68}x_{69} + 2x_{69} + x_{81}$$

$$R_{67} = -x_{10}x_{69} - x_{14} + x_{82}x_{83} + x_{84}$$

$$R_{68} = -1 + 2x_2 - 2x_2x_9 - 2x_2x_{20} + x_{82}$$

$$R_{69} = -x_9x_{220} - x_{13} + x_{83} + x_{220}$$

$$R_{70} = -6x_9x_{69} - 8x_{10}x_{69} + x_{10}x_{220} + x_{13} - x_{14} - 2x_{20}x_{69} + x_{85}$$

$$R_{71} = -x_{13}x_{70} + x_{14}x_{70} + x_{86}$$

$$R_{72} = -x_{71}x_{86} + x_{87}$$

$$R_{73} = -x_{72}x_{87} + x_{88}$$

$$R_{74} = -x_{73}x_{88} + x_{89}$$

$$R_{75} = -x_{74}x_{89} + x_{90}$$

$$R_{76} = -x_{75}x_{90} + x_{91}$$

$$R_{77} = -x_{76}x_{91} + x_{92}$$

$$R_{78} = -x_{77}x_{92} + x_{93}$$

$$R_{79} = -x_{84}x_{93} + x_{94}$$

$$R_{80} = -x_{85}x_{94} + x_{95}$$

$$R_{81} = -x_{95}^2 + x_{96}$$

$$R_{82} = -x_5 - x_5x_{17} + x_{17} + x_{97}$$

$$R_{83} = -x_{15}x_{97} + x_{98}$$

$$\begin{aligned} R_{84} = {} & -2x_1^2 + 2x_1x_{13} + 2x_1x_{14} + 2x_1x_{98} + 2x_1x_{99} - x_{13}^2 - 2x_{13}x_{98} - x_{14}^2 - 2x_{14}x_{99} \\ & + x_{18}x_{101} - x_{80}x_{96} - x_{98}^2 - x_{99}^2 \end{aligned}$$

$$R_{85} = -x_5x_{16} + x_{16} - x_{16}x_{97} + x_{99}$$

$$\begin{aligned} R_{86} = {} & -9x_9 - 9x_{10} - 4x_{20} - 2x_{23} - 4x_{25} - 2x_{28} - 2x_{30} - 4x_{31} - 2x_{33} - 2x_{35} - 4x_{38} \\ & - 2x_{39} - 2x_{42} - 2x_{46} - 4x_{67} - 2x_{68} + x_{100} \end{aligned}$$

$$R_{87} = -x_{102}x_{105} + x_{105} + x_{107}$$

$$R_{88} = -x_4x_{128} + x_{103}x_{210} - x_{106} - x_{107}x_{128}$$

$$R_{89} = x_{108}x_{210} - x_{109}x_{131} - x_{110}$$

$$R_{90} = x_{111}x_{210} - x_{112}x_{131} - x_{113}$$

$$R_{91} = x_{114}x_{210} - x_{115}x_{131} - x_{116}$$

$$R_{92} = x_{117}x_{210} - x_{118}x_{131} - x_{119}$$

$$R_{93} = x_{120}x_{211} - x_{122}x_{131} - x_{123}$$

$$R_{94} = x_{124}x_{211} - x_{125}x_{131} - x_{126}$$

$$R_{95} = -x_{18}x_{128} + x_{131}$$

$$R_{96} = x_{127} - x_{129}x_{131} - x_{130}$$

$$R_{97} = -x_{131}x_{133} + x_{132} - x_{134}$$

$$R_{98} = -x_{131}x_{136} + x_{135}x_{210} - x_{137}$$

$$R_{99} = -x_{131}x_{139} + x_{138}x_{210} - x_{140}$$

$$R_{100} = x_{141} - 2x_{141}x_{143} + x_{144}$$

$$R_{101} = -x_{141}x_{144} - x_{144} + x_{145}$$

$$R_{102} = 2x_{142} + x_{142}^2 + x_{144}^2 - x_{145}^2$$

$$R_{103} = -x_{142}x_{147} - x_{147} + x_{148}$$

$$R_{104} = -1 + x_{146}^2 + 2x_{146}x_{148} + x_{147}^2$$

$$R_{105} = x_3^2 - 2x_3x_{142} - 2x_{142} + x_{149}$$

$$R_{106} = -x_3x_{147} + x_5 - x_{146} - x_{147} + x_{149}x_{150}$$

$$R_{107} = 1 - 2x_4 + x_4^2 - 2x_4x_{142} + x_{151}$$

$$R_{108} = -x_4x_{183} + x_{18} + x_{151}x_{152} - x_{182}$$

$$R_{109} = -1 + 2x_{142}x_{194} - x_{153} + 2x_{194} - x_{194}^2$$

$$R_{110} = -1 + x_{102} + x_{153}x_{154} - x_{182} - x_{183}x_{194}$$

$$R_{111} = -x_{142}x_{156} - x_{156} + x_{157}$$

$$R_{112} = -1 + x_{155}^2 + 2x_{155}x_{157} + x_{156}^2$$

$$R_{113} = 1 + x_{142} - x_{142}x_{159} - x_{159} + x_{160}$$

$$R_{114} = x_{158}^2 + 2x_{158}x_{160} - 2x_{159} + x_{159}^2$$

$$R_{115} = -x_{142}x_{162} - x_{162} + x_{163}$$

$$R_{116} = -1 + x_{161}^2 + 2x_{161}x_{163} + x_{162}^2$$

$$R_{117} = -x_{142}x_{165} - x_{165} + x_{166}$$

$$R_{118} = -1 + x_{164}^2 + 2x_{164}x_{166} + x_{165}^2$$

$$R_{119} = -x_{142}x_{168} - x_{168} + x_{169}$$

$$R_{120} = -1 + x_{167}^2 + 2x_{167}x_{169} + x_{168}^2$$

$$R_{121} = -x_{142}x_{171} - x_{171} + x_{172}$$

$$R_{122} = -1 + x_{170}^2 + 2x_{170}x_{172} + x_{171}^2$$

$$R_{123} = -x_{142}x_{174} - x_{174} + x_{175}$$

$$R_{124} = -1 + x_{173}^2 + 2x_{173}x_{175} + x_{174}^2$$

$$R_{125} = -x_{142}x_{177} - x_{177} + x_{178}$$

$$R_{126} = -1 + x_{176}^2 + 2x_{176}x_{178} + x_{177}^2$$

$$R_{127} = -x_{142}x_{180} - x_{180} + x_{181}$$

$$R_{128} = -1 + x_{179}^2 + 2x_{179}x_{181} + x_{180}^2$$

$$R_{129} = 1 - x_{142}x_{183} - x_{183} + x_{184}$$

$$R_{130} = -1 + 2x_{182} + x_{182}^2 + 2x_{182}x_{184} + x_{183}^2$$

$$R_{131} = -x_{142}x_{186} - x_{186} + x_{187}$$

$$R_{132} = -1 + x_{185}^2 + 2x_{185}x_{187} + x_{186}^2$$

$$R_{133} = -x_{142}x_{189} - x_{189} + x_{190}$$

$$R_{134} = -1 + x_{188}^2 + 2x_{188}x_{190} + x_{189}^2$$

$$R_{135} = -x_{142}x_{192} - x_{192} + x_{193}$$

$$R_{136} = -1 + x_{191}^2 + 2x_{191}x_{193} + x_{192}^2$$

$$\begin{aligned} R_{137} = {} & -6 - 2x_0 - 3x_0^2 - 2x_0x_3 - x_1^2 - 2x_3 - 2x_3^2 - x_4^2 - 2x_5^2 + 2x_9 - x_9^2 - x_{10}^2 \\ & - x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{14}^2 - x_{15}^2 - x_{16}^2 - x_{17}^2 - x_{18}^2 - x_{100}^2 + 2x_{102} - 2x_{102}^2 \\ & - x_{103}^2 - x_{104}^2 - x_{108}^2 - x_{111}^2 - x_{114}^2 - x_{117}^2 - x_{120}^2 - x_{121}^2 - x_{124}^2 - x_{127}^2 \\ & - x_{128}^2 - x_{132}^2 - x_{135}^2 - x_{138}^2 + x_{141} - x_{194}^2 \end{aligned}$$

$$\begin{aligned} R_{138} = {} & x_4 - 2x_4x_9 - 2x_4x_{10} - 2x_4x_{11} - 2x_4x_{12} - 2x_4x_{13} - 2x_4x_{14} - 2x_4x_{15} \\ & - 2x_4x_{16} - 2x_4x_{17} + x_{194} \end{aligned}$$

$$R_{139} = -16 + x_0 - x_{195}^2$$

$$R_{140} = -x_{142}x_{197} - x_{197} + x_{206}$$

$$R_{141} = -1 + x_{196}^2 + x_{197}^2 - x_{206}^2$$

$$R_{142} = -x_{199} - x_{199}x_{200} + x_{207}$$

$$R_{143} = -1 + x_{198}^2 + x_{199}^2 - x_{207}^2$$

$$R_{144} = -x_{192}^2 + x_{208}$$

$$R_{145} = x_{197} - 2x_{202}x_{208} - x_{208}$$

$$R_{146} = -4x_{197} - 8x_{197}x_{201} + x_{200}$$

$$R_{147} = -x_{142} - x_{196}x_{203} + x_{200}$$

$$R_{148} = -x_{191} - x_{193} - x_{196}x_{204} + x_{198}$$

$$R_{149} = -x_{100} - 4x_{192}x_{205} + x_{199}$$

$$\begin{aligned} R_{150} = {} & 26 + 2x_0^2 + 2x_0x_3 - 2x_0x_{147} - 2x_0x_{156} + x_1^2 - 2x_1x_{183} + x_3^2 - 2x_3x_{156} + x_5^2 \\ & - 2x_5x_{174} + x_9^2 - 2x_9x_{159} + x_{10}^2 - 2x_{10}x_{162} + x_{11}^2 - 2x_{11}x_{165} + x_{12}^2 \\ & - 2x_{12}x_{168} + x_{13}^2 - 2x_{13}x_{171} + x_{14}^2 - 2x_{14}x_{177} + x_{15}^2 - 2x_{15}x_{180} + x_{16}^2 \\ & - 2x_{16}x_{186} + x_{17}^2 - 2x_{17}x_{189} + x_{100}^2 - 2x_{100}x_{192} - x_{142}x_{209} - 2x_{146} - 2x_{148} \\ & + x_{148}^2 - 2x_{155} - 2x_{157} + x_{157}^2 - 2x_{158} + 2x_{159} - 2x_{160} + x_{160}^2 - 2x_{161} - 2x_{163} \\ & + x_{163}^2 - 2x_{164} - 2x_{166} + x_{166}^2 - 2x_{167} - 2x_{169} + x_{169}^2 - 2x_{170} - 2x_{172} + x_{172}^2 \\ & - 2x_{173} - 2x_{175} + x_{175}^2 - 2x_{176} - 2x_{178} + x_{178}^2 - 2x_{179} - 2x_{181} + x_{181}^2 - 2x_{182} \\ & + x_{184}^2 - 2x_{185} - 2x_{187} + x_{187}^2 - 2x_{188} - 2x_{190} + x_{190}^2 - 2x_{191} - 2x_{193} + x_{193}^2 \end{aligned}$$

$$R_{151} = -x_{104}x_{210} + x_{128}$$

$$R_{152} = -x_{121}x_{211} + x_{128}$$

$$\begin{aligned} R_{153} = & -2 - x_{100}^2 - x_{147}^2 - x_{156}^2 + 2x_{159} - x_{159}^2 - x_{162}^2 - x_{165}^2 - x_{168}^2 - x_{171}^2 - x_{174}^2 \\ & - x_{177}^2 - x_{180}^2 - x_{183}^2 - x_{186}^2 - x_{189}^2 + x_{192}^2 - x_{212}^2 - x_{213}^2 - x_{214} - x_{214}^2 \end{aligned}$$

$$\begin{aligned} R_{154} = & -1 - 4x_{106}^2 - 4x_{110}^2 - 4x_{113}^2 - 4x_{116}^2 - 4x_{119}^2 - 4x_{123}^2 - 4x_{126}^2 + x_{128}^2 - 4x_{130}^2 \\ & - 4x_{134}^2 - 4x_{137}^2 - 4x_{140}^2 - x_{215}^2 - x_{216}^2 - x_{217} - x_{217}^2 - 4x_{219}^2 \end{aligned}$$

$$R_{155} = -x_4x_{218} + x_{128}$$

$$R_{156} = x_2x_{128} - x_5x_{218} - x_{219}$$

$$R_{157} = -x_2x_{13} + x_{220}$$

$$\begin{aligned} R_{158} = & x_{102}x_{222} - 2x_{103}x_{156} + 2x_{108} - 2x_{108}x_{159} - 2x_{111}x_{162} - 2x_{114}x_{165} - 2x_{117}x_{168} \\ & - 2x_{120}x_{171} - 2x_{124}x_{177} - 2x_{127}x_{180} - 2x_{132}x_{186} - 2x_{135}x_{189} - 2x_{138}x_{192} \\ & - 2x_{142}x_{222} + 2x_{155}x_{156} + 2x_{156}x_{157} - 2x_{158} + 2x_{158}x_{159} + 2x_{159}x_{160} - 2x_{160} \\ & + 2x_{161}x_{162} + 2x_{162}x_{163} + 2x_{164}x_{165} + 2x_{165}x_{166} + 2x_{167}x_{168} + 2x_{168}x_{169} \\ & + 2x_{170}x_{171} + 2x_{171}x_{172} + 2x_{176}x_{177} + 2x_{177}x_{178} + 2x_{179}x_{180} + 2x_{180}x_{181} \\ & + 2x_{185}x_{186} + 2x_{186}x_{187} + 2x_{188}x_{189} + 2x_{189}x_{190} + 2x_{191}x_{192} + 2x_{192}x_{193} + x_{221} \\ & - 2x_{222} \end{aligned}$$

$$\begin{aligned} R_{159} = & -1 - x_{102}x_{221} + x_{103}^2 - 2x_{103}x_{155} + x_{108}^2 - 2x_{108}x_{158} + x_{111}^2 - 2x_{111}x_{161} \\ & + x_{114}^2 - 2x_{114}x_{164} + x_{117}^2 - 2x_{117}x_{167} + x_{120}^2 - 2x_{120}x_{170} + x_{124}^2 - 2x_{124}x_{176} \\ & + x_{127}^2 - 2x_{127}x_{179} + x_{132}^2 - 2x_{132}x_{185} + x_{135}^2 - 2x_{135}x_{188} + x_{138}^2 - 2x_{138}x_{191} \\ & + x_{155}^2 - x_{156}^2 + x_{158}^2 + 2x_{159} - x_{159}^2 + x_{161}^2 - x_{162}^2 + x_{164}^2 - x_{165}^2 + x_{167}^2 \\ & - x_{168}^2 + x_{170}^2 - x_{171}^2 + x_{176}^2 - x_{177}^2 + x_{179}^2 - x_{180}^2 + x_{185}^2 - x_{186}^2 + x_{188}^2 \\ & - x_{189}^2 + x_{191}^2 - x_{192}^2 + x_{222} \end{aligned}$$

$$\begin{aligned} R_{160} = & x_{102}x_{224} - 2x_{104}x_{147} - 2x_{121}x_{174} - 2x_{128}x_{183} - 2x_{142}x_{224} + 2x_{146}x_{147} \\ & + 2x_{147}x_{148} + 2x_{173}x_{174} + 2x_{174}x_{175} + 2x_{182}x_{183} + 2x_{183} + 2x_{183}x_{184} + x_{223} \\ & - 3x_{224} \end{aligned}$$

$$\begin{aligned} R_{161} = & -x_{102}x_{223} + x_{104}^2 - 2x_{104}x_{146} + x_{121}^2 - 2x_{121}x_{173} + x_{128}^2 - 2x_{128}x_{182} + x_{146}^2 \\ & - x_{147}^2 + x_{173}^2 - x_{174}^2 + x_{182}^2 - x_{183}^2 + x_{223} + x_{224} \end{aligned}$$

B Artifact identities

The repository snapshot is

`a0ef2df5bb1f73c35cd334bbc7f53553350ef06a`.

The SHA-256 identities below fix the literal input and output.

Pinned 245-variable parent

`1a42dd3a148c3084cd95a367869d251ed92e2f9852397215e4909e70b03d278e`

Final 225-variable residual specification

`fee84c408828aba3e7ceebe9a37f58509327063aa7fd078fd49c80adf3af4e56`

Collected 7,867-term polynomial

`3a9f582c1be2c4e4f927b383ffa66eb831a5a018c1f5d8441a16f58251ae14b0`

Final metrics

`b3f4606cf084a48a0652f3c3130e43adefecc8b76df1f4ae3a150033ee9cc054`

The package-wide `SHA256SUMS` authenticates the remaining files, including this report after PDF compilation. `provenance/SOURCES.json` records the source origins and exact paths. Hashes identify artifacts; their agreement does not replace the mathematical proof.